Standard Service Procedures

Generic, manufacturer-agnostic procedures for common maintenance tasks. Always confirm torque specifications, fluid types, and part numbers against the vehicle-specific manual before performing any of these procedures — the steps below describe the general process, not brand-specific specifications.

Engine Oil & Filter Change

Tools needed: Oil filter wrench, drain pan, socket set, funnel, jack and jack stands (if needed)

Procedure:

1. Warm the engine for 2-3 minutes so oil flows more freely, then shut it off.
2. Raise and safely support the vehicle if the drain plug is not accessible from below.
3. Place the drain pan under the oil pan drain plug, remove the plug, and allow oil to fully drain (5-10 minutes).
4. Remove the old oil filter, allowing residual oil to drain into the pan.
5. Lightly coat the new filter's gasket with fresh oil and hand-tighten it into place — do not over-tighten.
6. Reinstall the drain plug with a new sealing washer if required, torqued to the manufacturer's specification.
7. Refill with the specified oil grade and quantity, then check the level with the dipstick after a few minutes.
8. Start the engine, check for leaks around the filter and drain plug, then recheck the oil level once more when cool.

Notes: Dispose of used oil and filters at an approved recycling facility — never pour used oil down a drain.

Brake Pad Inspection & Replacement

Tools needed: Lug wrench, jack and jack stands, C-clamp or brake piston tool, wire brush

Procedure:

1. Loosen lug nuts slightly, then raise and support the vehicle securely.
2. Remove the wheel to expose the brake caliper and pads.
3. Inspect pad thickness; if the friction material measures below approximately 3mm, replace the pads.
4. Remove the caliper (typically two bolts) and support it so it does not hang from the brake hose.
5. Remove the old pads and clean the caliper bracket and slide pins of rust and debris.
6. Compress the caliper piston back into its bore to make room for the new, thicker pads.
7. Install the new pads, reposition the caliper, and torque the caliper bolts to specification.
8. Reinstall the wheel, torque lug nuts to specification, and pump the brake pedal several times before driving to seat the pads.

Notes: Replace pads in axle pairs (both fronts or both rears together) to maintain even braking. If rotors are scored or below minimum thickness, resurface or replace them at the same time.

Battery Testing & Replacement

Tools needed: Digital multimeter, battery terminal cleaner/wire brush, wrench for terminal clamps, safety glasses and gloves

Procedure:

1. With the engine off and vehicle rested for at least an hour, measure resting voltage across the terminals; expect roughly 12.6V for a healthy, fully-charged battery.
2. Start the engine and measure voltage again; expect roughly 13.8-14.7V, confirming the charging system is functioning.
3. If replacing the battery, disconnect the negative terminal first, then the positive terminal, to avoid short circuits.
4. Remove the battery hold-down clamp and lift the battery out.
5. Clean the tray and terminal clamps of any corrosion before installing the new battery.
6. Install the new battery, secure the hold-down clamp, then reconnect the positive terminal first, followed by the negative.
7. Verify resting and running voltage on the new battery match expected ranges.

Notes: Always disconnect negative before positive when removing a battery, and reconnect in the reverse order, to minimize the risk of a short circuit from a tool bridging the terminals.

Coolant System Flush

Tools needed: Drain pan, funnel, coolant of the manufacturer-specified type, distilled water

Procedure:

1. Allow the engine to cool completely — never open a cooling system while the engine is hot.
2. Place a drain pan under the radiator drain valve or remove the lower radiator hose to drain old coolant.
3. Close the drain valve/reconnect the hose, then fill the system with a flushing solution or distilled water and run the engine to operating temperature with the heater on full.
4. Drain the flush solution completely, then repeat with plain distilled water until the drained liquid runs clear.
5. Refill with the manufacturer-specified coolant type, mixed to the correct ratio with distilled water.
6. Run the engine with the cap off (or in the fill position) to bleed air pockets, topping up as the level drops, then seal the system.
7. Recheck coolant level once the engine has fully cooled, topping up the overflow reservoir as needed.

Notes: Never mix coolant types (e.g. OAT and HOAT formulations) unless the container confirms compatibility — mixing incompatible coolants can cause gel formation that blocks the cooling system.

Air Filter Replacement

Tools needed: Screwdriver or clips depending on housing type

Procedure:

1. Locate the air filter housing, typically a large plastic box connected to the intake tube.
2. Release the clips or screws securing the housing lid.
3. Remove the old filter, noting its orientation.
4. Wipe out any debris from inside the housing.
5. Install the new filter in the same orientation as the old one.
6. Reseal the housing lid and clips/screws.

Notes: A visibly dirty air filter can measurably reduce fuel economy and acceleration response; most manufacturers recommend inspection every 12,000-15,000 km.